Tutorial 2

Introduction To Programming

Solutions

Q1.

Given n integers and an integer x, we want to make every number in the list equal to x. In one step we can either increase or decrease an element in the list by 2. Find the minimum number of steps required to make every element in the list equal to x or tell if it is impossible to do so.

Solution

```python
## Tut 2
n= int(input())
## n  taken as input
x = int(input())
## x has been taken as input
steps = 0
isPossible = True ## a boolean variable to check whether
                  ## all the elements can reach x by adding or subtracting
                  ## in steps of two
for i in range(n):
    ## all the elements from 1 to n
    ## will be taken as input inside the loop
    y = int(input())
    if y > x :
      ## y is greater than x
      if (y - x) % 2 !=0:
        ## y can be decremented in steps of two
        isPossible = False
      else :
        steps += (y - x) // 2
    elif y < x:
      if (x - y) % 2 !=0:
        ## y can be incremented in steps of two
        isPossible = False
      else:
        steps += (x - y) // 2
if isPossible:
  print(steps)
else :
  print("Not Possible")
```

Input:

```
4
3
1
5
7
9
```

Output

```
7
```

Q2.

Given **n** integers, find the sum of the elements of the list excluding the segments that start with 10 and ends with 20. It is guaranteed that only one segment is present that starts with 10 and ends with 20.

Solution

```python
total = 0
## total initalized with 0

take = True
## a boolean set to true
## this will help us to exclude the desired segment
n = int(input())
for i in range(n):
   x = int(input())
   if x == 10:
       take = False
   elif x == 20 :
       take  = True
   elif take:
       total += x
print(total)
```

Input:

```
6
2
16
10
3
20
9
```

Output

```
27
```